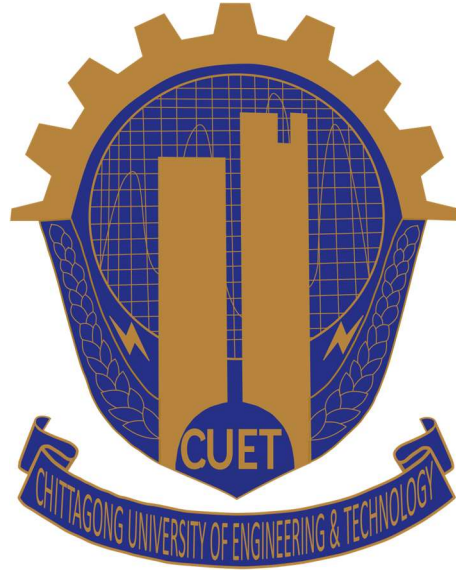




Chittagong University of Engineering and Technology (CUET)

Department of Biomedical Engineering

Project Name: Price Scraping of Medicines Based on Disease Input

Course Name: Computer Programming and Electrical Analysis Sessional

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Abstract:

The Project is about Scraping Medicine Prices from Different Websites and Generating Notifications for Antibiotics in Python.

By using this code user can input his/her disease's name and the programme will show the suitable medicine for the disease. It also scrapes medicine with different brands. Also, the Scraper will scrape the price and show it in descending order with a clear graph. When the user inputs a disease that requires antibiotics, the code will give an alarm with a notification that it is an Antibiotic and Doctor permission is needed. We have used different types of user-defined functions and built-in functions. Loops and dictionaries have been used, also python module Request, bs4, and Panda. Matplotlib is used for graph plotting. BeautifulSoup Python library used for pulling data out of HTML and XML files

Objective:

- Take symptoms as input and suggest expert medication in the blink of an eye.
- Compare the price and availability of the suggested medication to provide easy healthcare for various people of diverse socio-economic conditions.
- Notify patients if any antibiotics are needed.
- Provide a user-friendly graphical interface.

Introduction:

We developed a Python program that will come with a graphical user interface. It will collect common symptoms as user input, and prescribe medications. By web scraping (the act of going through web pages and extracting selected text or images) our program will also display the market price and availability of the medicines. If a major injury or infection is detected and the patient needs to take antibiotics, it will also notify the user.

Our program will provide a user-friendly environment and take data from users to check health issues.

On input: Users will use some common keywords (e.g., *fever, burn, cut, acidity* etc.)

On output:

1. It will suggest common medications and diagnostic approaches.
2. Compare the price of prescribed medicine with different brands and show a graphical view and availability to easily afford medicines for all types of consumers in a complex socio-economic system.
3. The program will notify the user if any Antibiotic is needed for medication

To program the code, we used **Python-3** and used some of its basic functions and operations. Such as:

1. Loop
2. Function
3. Dictionary (Define)

Imported Functions:

1. Time
2. Plyer

3. re.search module

For Scraping, we have used:

1. Request
2. BeautifulSoup

In order **to plot** we used:

1. Matplotlib

Pandas is used for creating data frames.

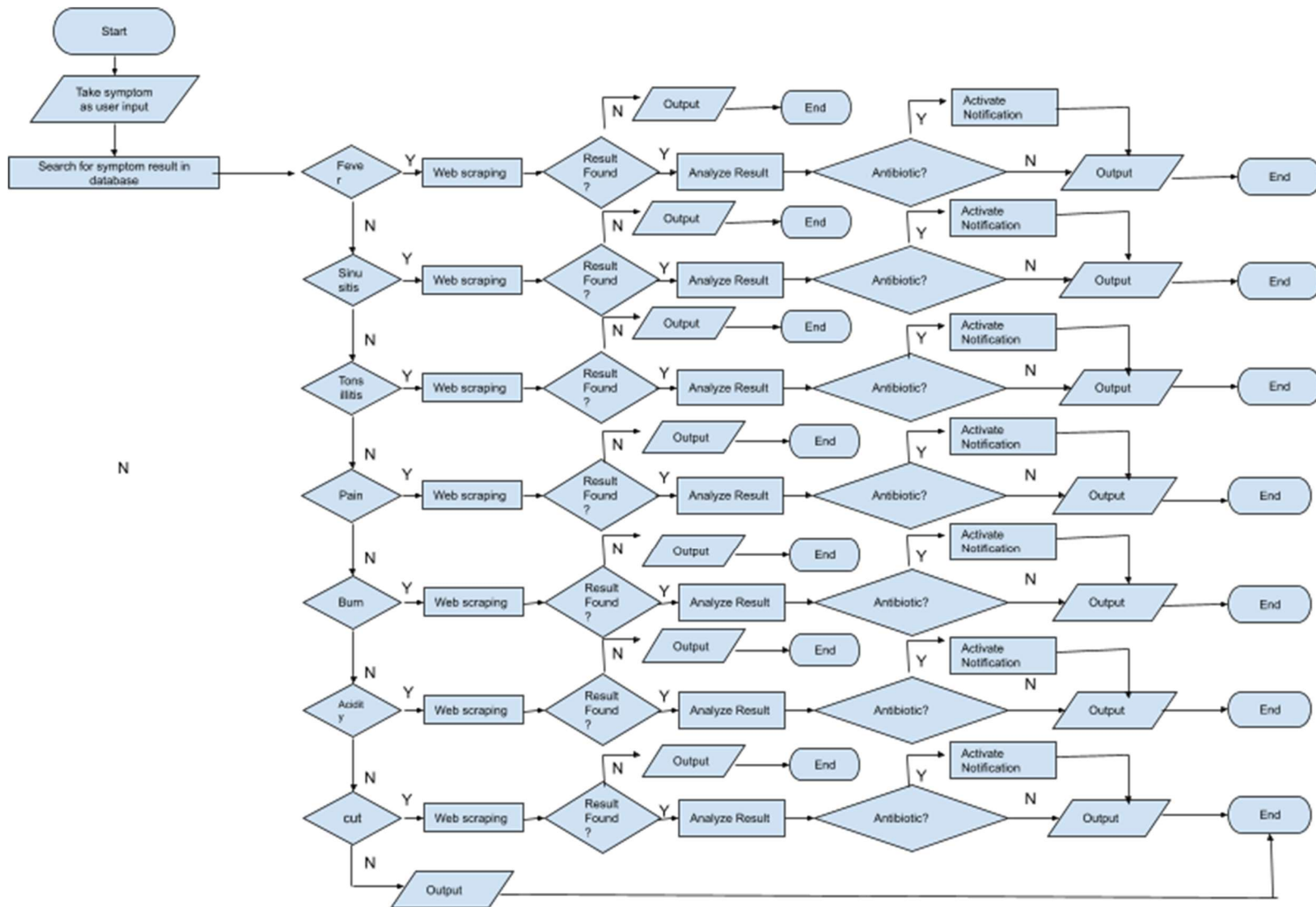
Motivation:

Our goal is to create healthcare technology for a brighter future as future biomedical engineers. Fever affects an adult individual on average about 2–3 times a year^[1]. Children endure more pain. A sore throat affects 7.5% of people every three months, and 2% of people consult a doctor for tonsillitis every year^[2]. Because of their economic and social circumstances, people frequently avoid going to the doctor. We used Python to try to solve this issue. Everybody will be able to access expert medication for free and conveniently.

We chose this project because this kind of work is still limited, and this project has huge potential in the future. As this code is easy to understand, we can add many other features in future work.

Implementation:

❖ Flowchart:



❖ Data Structure:

Our program follows a **linear static data structure**. It has a fixed memory size and doesn't store data afterwards, making it a static structure. After that, our program terminates just after showing the result. We can't change its value during runtime.

❖ Functions:

- **User Defined Functions:** User-defined functions are tools we used to arrange our code within a policy's body. Once a function has been defined, it can be used in the same way that the built-in action and parser functions are used. Instead of being passed by value, variables are passed by reference to functions.

The User defined function we used are

- **Scrap:**

- ◆ We used a scrap function to scrape the website to find out the price, brand name, amount of dose, etc. of medicine on a website.

Syntax:

```
def scrap(aroggo_url):  
    statements  
    .  
    .  
    return {  
        'medicine_name':medicine_name,  
        'meidicine_price':meidicine_price,  
    }
```

- **myplot:**

- ◆ to plot graphical values to make a user-friendly experience. Here we call a function **matplotlib.pyplot** to conduct operations in Python.

Syntax:

```
def myPlot(name,price):  
    statements
```

```
    .  
    .
```

```
myPlot(medicine_name,medicine_price)
```

➤ **Built-in Functions:**

- **float():** Returns a floating point number
- **print():** Prints to the standard output device
- **import:** Python code in one module can access Python code in another module by importing the other module's code.
 - **request:** To get into the URL and request access to a website for scraping. Requests library handling the interaction with the web page (Using HTTP requests)

Syntax:

```
import requests
```

```
page = requests.get(url=aoggo_url)
```

- **from bs4 import BeautifulSoup:** Simple techniques for navigating, searching, and altering a parse tree in HTML and XML files are provided by BeautifulSoup.

It converts a complicated HTML document into a Python object tree.

Syntax:

```
from bs4 import BeautifulSoup

page = requests.get(url=aoggo_url)

soup = BeautifulSoup(page.content, 'html.parser')

price_element = soup.find(class_ = 'js9')
```

- ***plyer()***: Plyer is a Python platform-independent API for using features frequently present on multiple platforms, particularly mobile ones.

Syntax:

```
from plyer import notification

for name in medicine_name:

    if (found):

        notification.notify(

            title = "**Antibiotic Detected!!!",

            message = "Take Medicine Carefully ",

        )
```

- ***time()***: The time is returned by the Python `time()` function as a floating point value in UTC that is represented in seconds since the epoch.

Syntax:

```
import time
# displaying time
time.sleep(value)
# waiting time
time.sleep(value)
```

- ***matplotlib.pyplot()***: is a state-based interface to matplotlib. It provides an implicit, MATLAB-like, way of plotting. It also opens figures on your screen, and acts as the figure GUI manager.

Syntax:

```
import matplotlib.pyplot as plt

plt.plot(xpoints, ypoints)

plt.show()
```

- ***dict()***: function creates a dictionary. A dictionary is a collection which is unordered, changeable and indexed. Dictionary items are ordered, changeable, and does not allow duplicates. Dictionary items are presented in key:value pairs, and can be referred to by using the key name.

- ***input()***: To collect user input in Python, we use the `input()` function. The input function converts the input we provide into a string. The `input()` function will transform an integer number we enter, into a string.

Syntax:

```
input_value=input("enter the problem=")
```

- ***len()***: The `len()` function **returns the number of items in an object**. When the object is a string, the `len()` function returns the number of characters in the string.

Syntax:

```
result_length=len(result_data)
```

- ***str()***: function converts values to a string form so they can be combined with other strings.

➤ ***Special Library Functions:***

- ***Pandas***: Pandas is a library used for data manipulation and analysis. It is used to extract the data and store it in the desired format.
 - ***pd.dataframe()***: A Pandas DataFrame is a two-dimensional data structure, similar to a two-dimensional array or a table with rows and columns.

Syntax:

```
dataset =  
pd.DataFrame(data,columns=columns)
```

➤ **Memory Usage:** Our program has a size of ** MB, and it stores all its data into the location:

C:\Users\NEHA\AppData\Local\Temp\ipykernel_9200\906433087

As the location is in a local storage instead of a cloud storage, it is an offline storage. The data is being handled by the CPU of the user. By the time a user uses the program more it will store data. In future we will develop our code more to work with medication history by using memory.

Discussion:

In this project, we took symptoms as input to suggest medication instantly and generate notifications for Antibiotics. In addition to that, we compared medicine prices and availability from different websites to give an overview to the user, using the Python programming language. The data collected in this project can be used for predicting the price of medicine for particular symptoms. There are some limitations of our programming code.

1. The main limitation of web scraping by our code is probably getting blocked by the website. It mainly depends on the size of the operation, but if we send out too many requests from a single IP address and the website has strict rules on web scraping, we can get the IP blocked by the website.
2. If the website holder changes the HTML code of the website, then the web scraping code will show errors as the element class will be changed.
3. As we web-scraped data directly from the website text, some of the times there is repeated power value of the same medicine (i.e., napa 500 mg 500 mg) which can't be stripped by outer code.
4. Our notification-generated code could not show the name of the particular antibiotic.
5. The upper limit for pandas Dataframe was 100 GB of free disk space on the machine, so we can't save so much symptom medicine as a CSV file.
6. In Panda for creating new rows, the use of append is deprecated. Using it row by row leads to very slow code.
7. We are still developing the code. After full development, our program will be able to address complex health issues and make remote healthcare systems more handy by connecting patients with an AI that is an expert like a doctor.

Conclusion:

Fever, tonsillitis, sinusitis, burns, etc are some common diseases widespread among people. Some people just don't want to go to the doctor for financial reasons or a busy schedule. With our project, it will be easy for them to find the medicine for their diseases, only by inputting the name of the illness, which is very easy to use and not time-consuming.

They can compare the price and company themselves by seeing the visual histogram. Our project is user-friendly and easy to understand. We can work with pharmaceuticals in the future in this project and add their medicines to this project, which is a huge possibility for marketing. Also, many features like suggesting medicines with the symptoms of the patients, Antibiotic alarms, etc can be added to make the program more sophisticated.

Appendix:

1. <https://www.cdc.gov/dotw/common-cold/index.html#:~:text=Each%20year%20in%20the%20United,and%20children%20have%20even%20more>.
2. <https://en.wikipedia.org/wiki/Tonsillitis#:~:text=About%207.5%25%20of%20people%20have,doctor%20for%20tonsillitis%20each%20yea>